Magnetism

Part 1

LEARNING OBJECTIVES

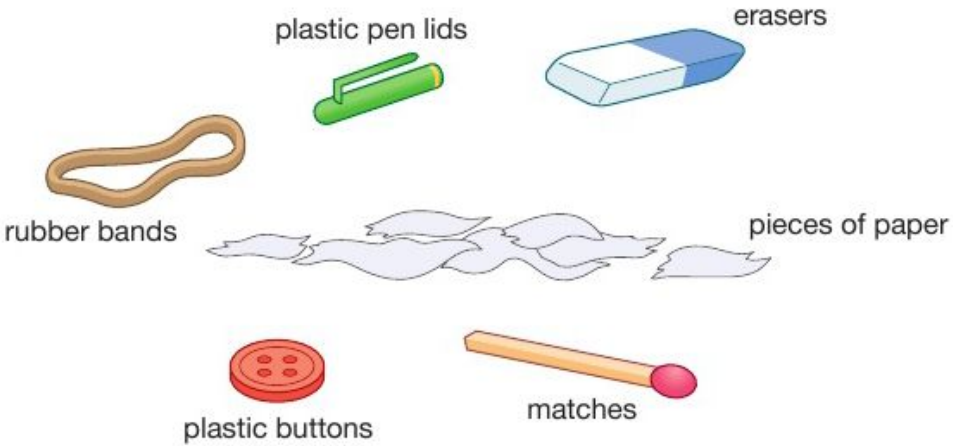
- Know that magnets repel and attract other magnets and attract magnetic substances
- Describe the properties of magnetically hard and soft materials
- Practical: investigate the magnetic field pattern for a permanent bar magnet and between two bar magnets
- Understand the term magnetic field line
- Know that magnetism is induced in some materials when they are placed in a magnetic field
- Describe how to use two permanent magnets to produce a uniform magnetic field pattern
- Know that an electric current in a conductor produces a magnetic field around it

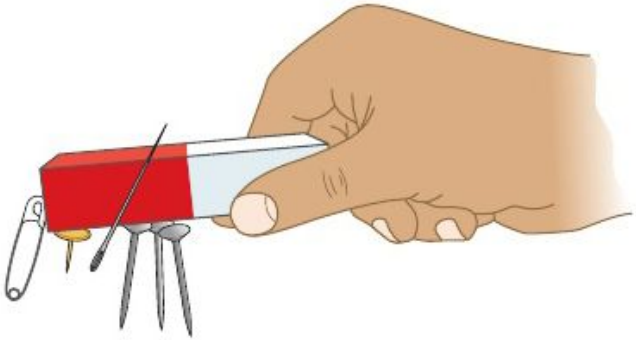
PHYSICS ONLY

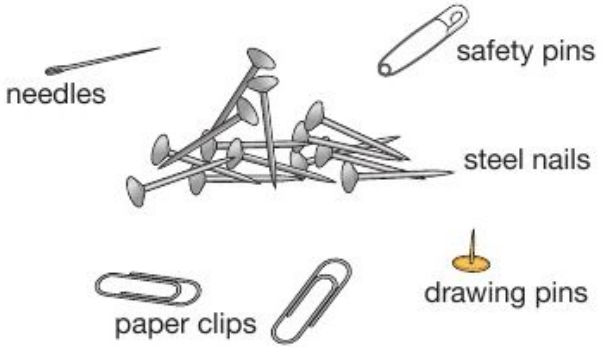
- Describe the construction of electromagnets
- Draw magnetic field patterns for a straight wire, a flat circular coil and a solenoid when each is carrying a current

MAGNETISM AND MAGNETIC MATERIALS

Magnets are able to attract objects made from magnetic materials such as iron, steel, nickel and cobalt. Magnets cannot attract objects made from materials such as plastic, wood, paper or rubber. These are non-magnetic materials.



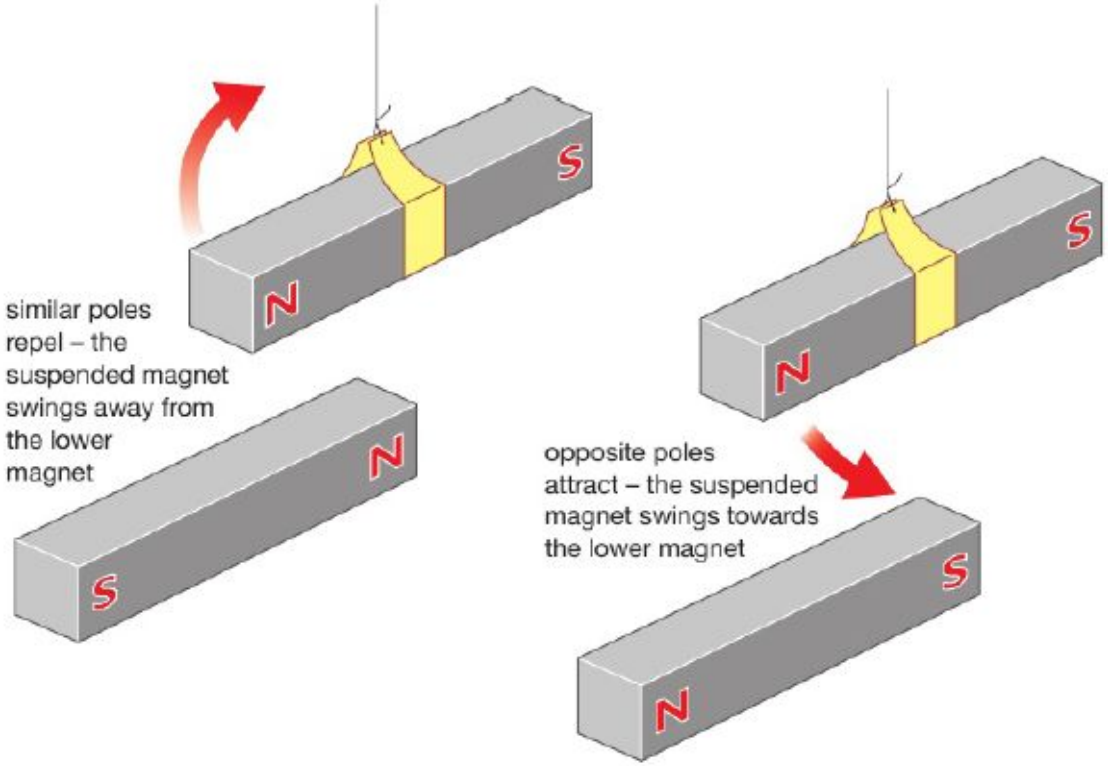




MAGNETS

The strongest parts of a magnet are called its poles. Most magnets have two poles. These are called the north pole and the south pole.

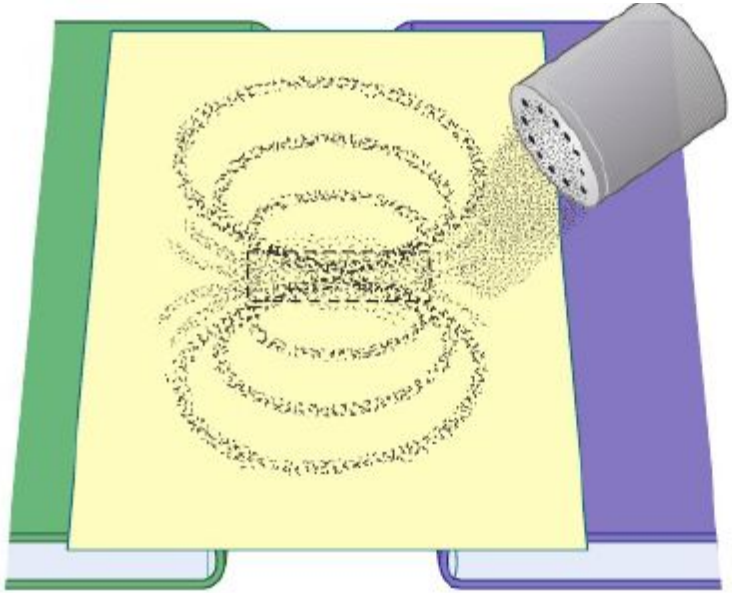
If two similar poles are placed near to each other they repel. If two dissimilar (opposite) poles are placed near to each other they attract.



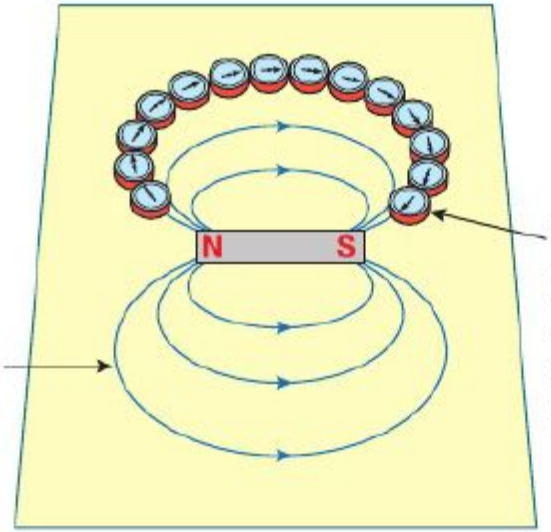
Permanent magnets like the bar magnets shown in Figure 20.3 are made from a magnetically hard material such as steel. A magnetically hard material keeps its magnetism once it has been magnetised. Iron is a magnetically soft material and is not suitable for a permanent magnet. Magnetically soft materials lose their magnetism easily and are therefore useful as temporary magnets.

MAGNETIC FIELDS

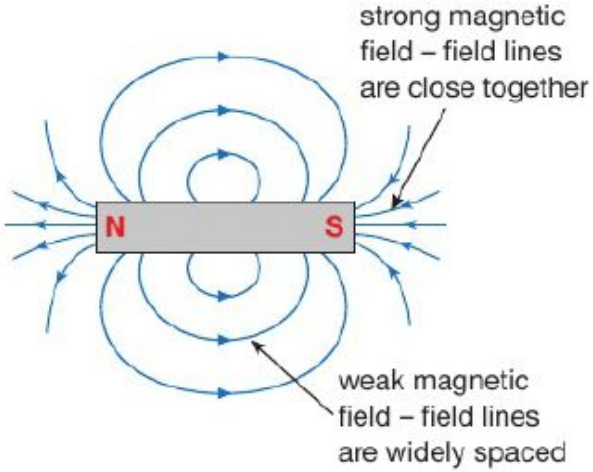
Around every magnet there is a volume of space where we can detect magnetism. This volume of space is called a **magnetic field**. Normally a magnetic field cannot be seen but we can use iron filings or plotting compasses to show its shape and discover something about its strength and direction.



if you make a pencil dot at the end of the needle in each position of the compass, you can plot the field line



in each position, the needle of the plotting compass lines up with the field line, pointing from north to south



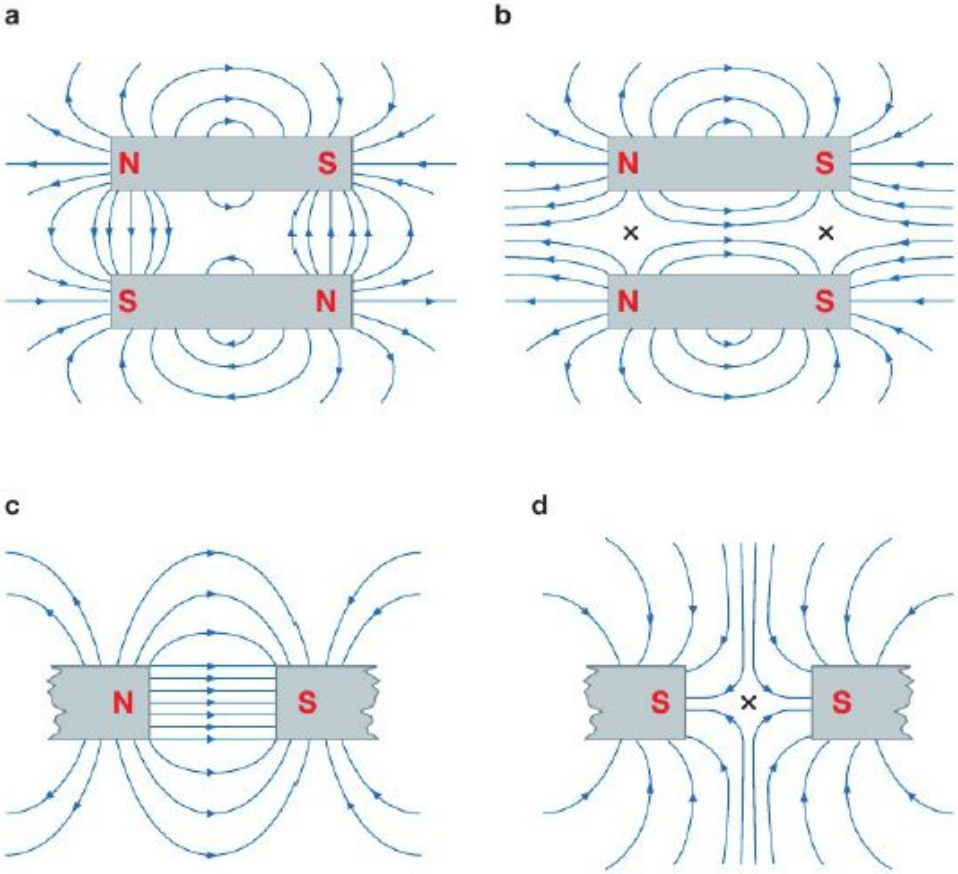
We draw magnetic fields like that in Figure 20.5 using magnetic field lines. Magnetic field lines don't really exist but they help us to visualise the main features of a magnetic field.

The magnetic field lines:

- show the shape of the magnetic field
- show the direction of the magnetic force – the field lines 'travel' from north to south
- show the strength of the magnetic field – the field lines are closest together where the magnetic field is strongest.

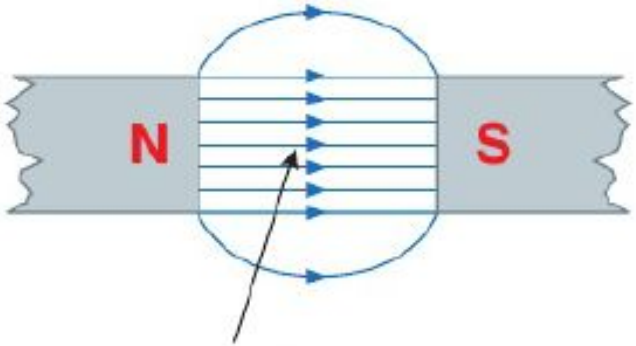
OVERLAPPING MAGNETIC FIELDS

If two magnets are placed near each other, their magnetic fields overlap and affect each other. We can investigate this using iron filings or plotting compasses. Figure 20.6 shows the different field patterns we would see.

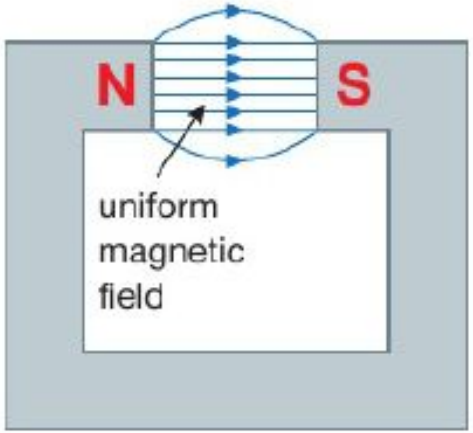


CREATING A UNIFORM MAGNETIC FIELD

If we look very carefully at the magnetic field created between the north pole of one magnet and the south pole of a different magnet we will see that the field is shown as a series of straight field lines that are evenly spaced. A field like this is described as a uniform magnetic field – that is, its strength and direction is the same everywhere. Creating a uniform field like this is extremely useful as we will see later in this chapter and in the next chapter.



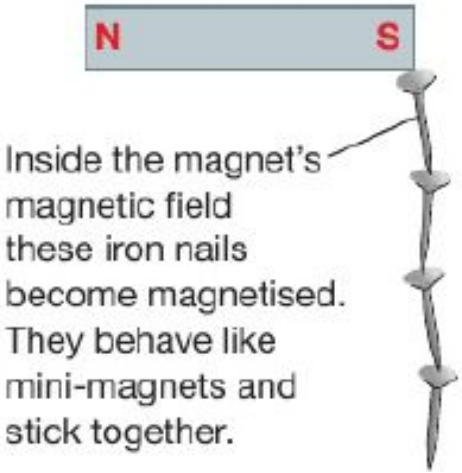
uniform field between the poles of the magnets

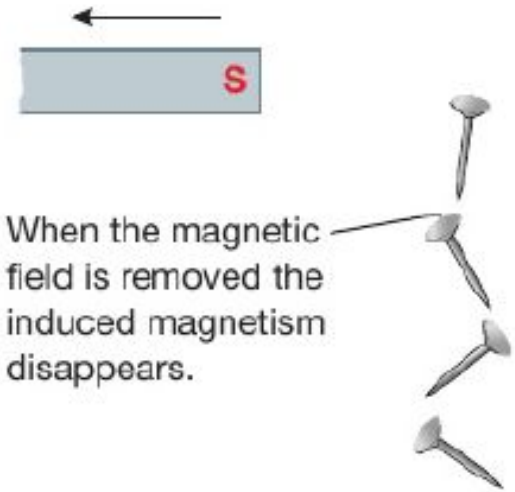


u-shaped permanent magnets can be used to create a uniform magnetic field

INDUCED MAGNETISM

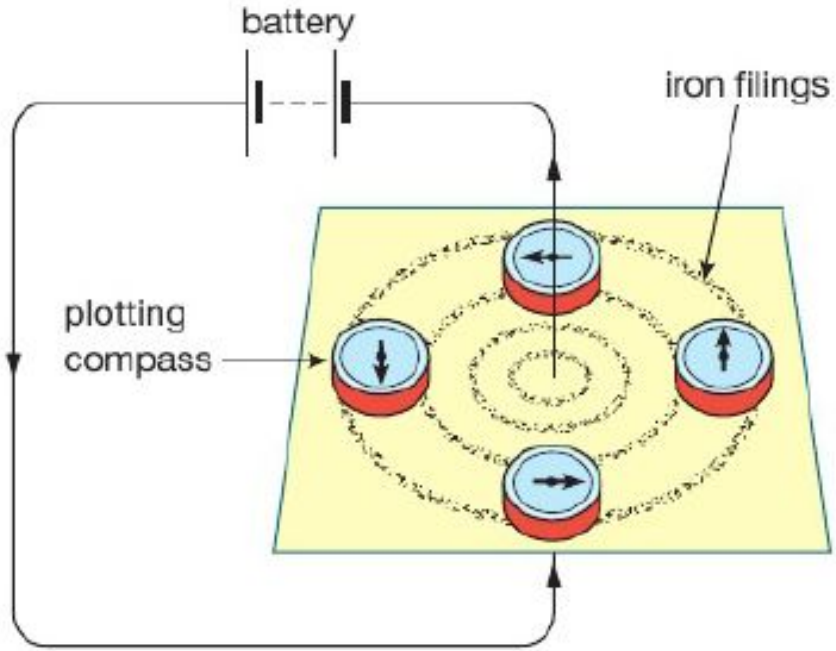
If we place an object made from a magnetic material, for example, an iron nail, inside a magnetic field it becomes a magnet. We say that magnetism has been induced in the nail. Because iron is a magnetically soft material, its induced magnetism is temporary and disappears if the permanent magnet is removed. If the nail is made from a magnetically hard material such as steel it will retain some of its magnetism after the magnet is removed.



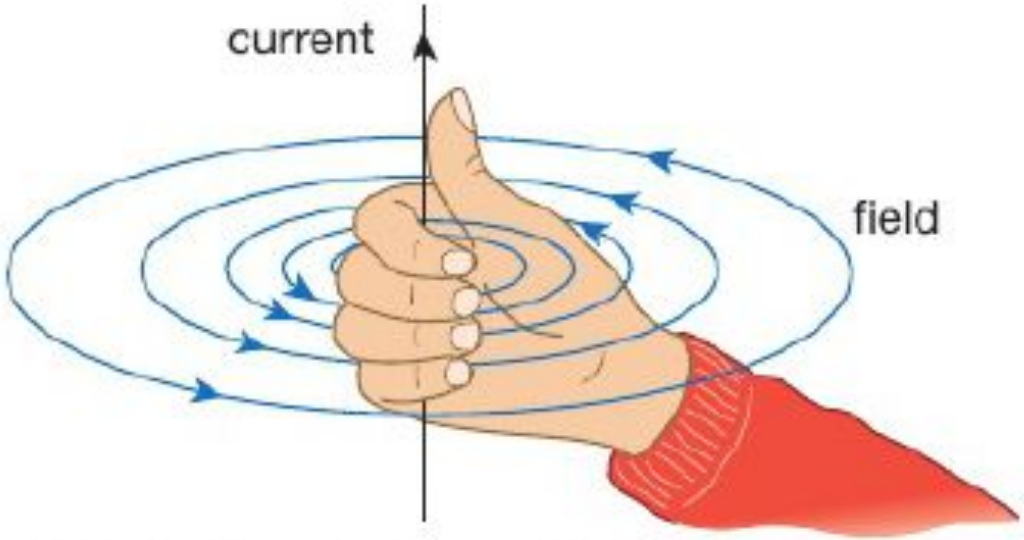


ELECTROMAGNETISM

When there is a current in a wire a magnetic field is created around it. This is called electromagnetism. The field around the wire is quite weak and circular in shape.



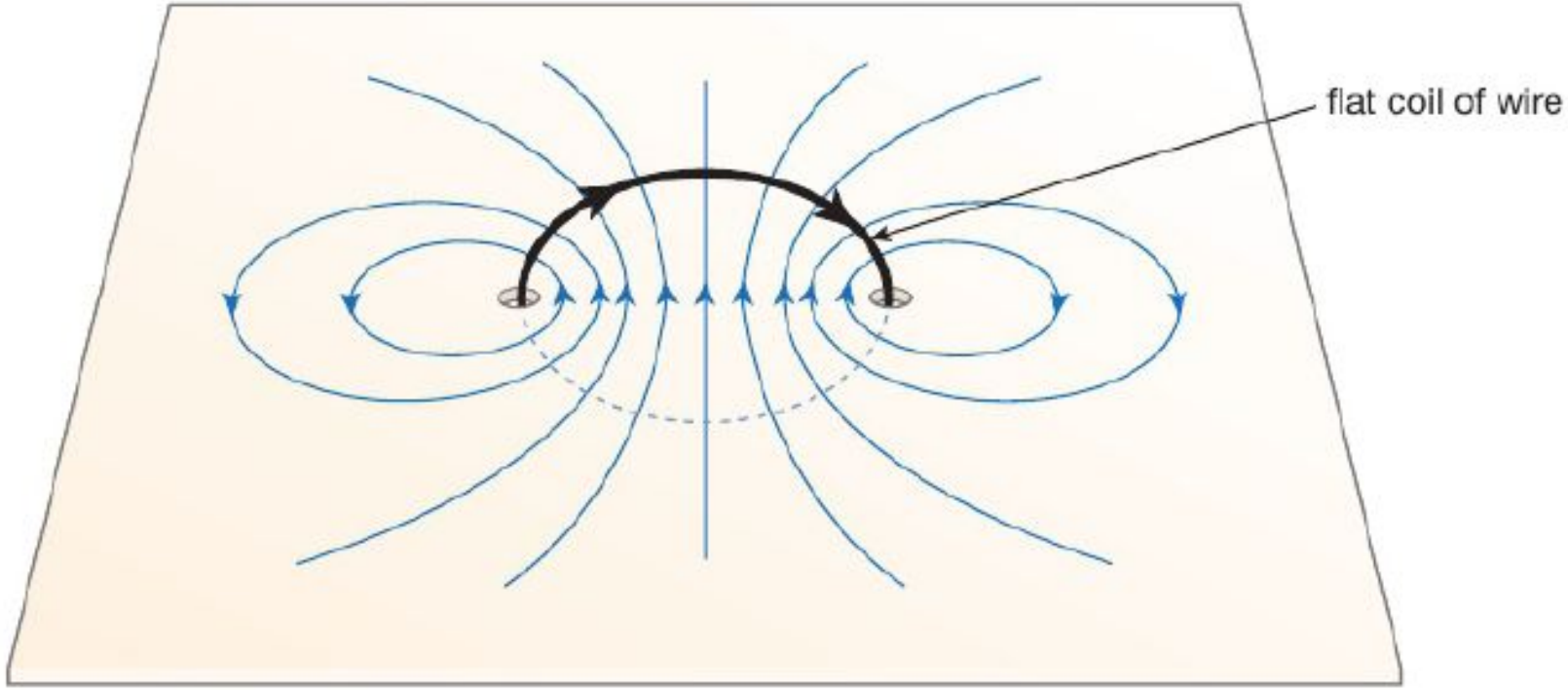
The shape and direction of the magnetic field around a current-carrying wire can be seen using iron filings and plotting compasses.
Changing the direction of the current, changes the direction of the magnetic field.



With the thumb of your right hand pointing in the direction of the current, your fingers will curl in the direction of the field.

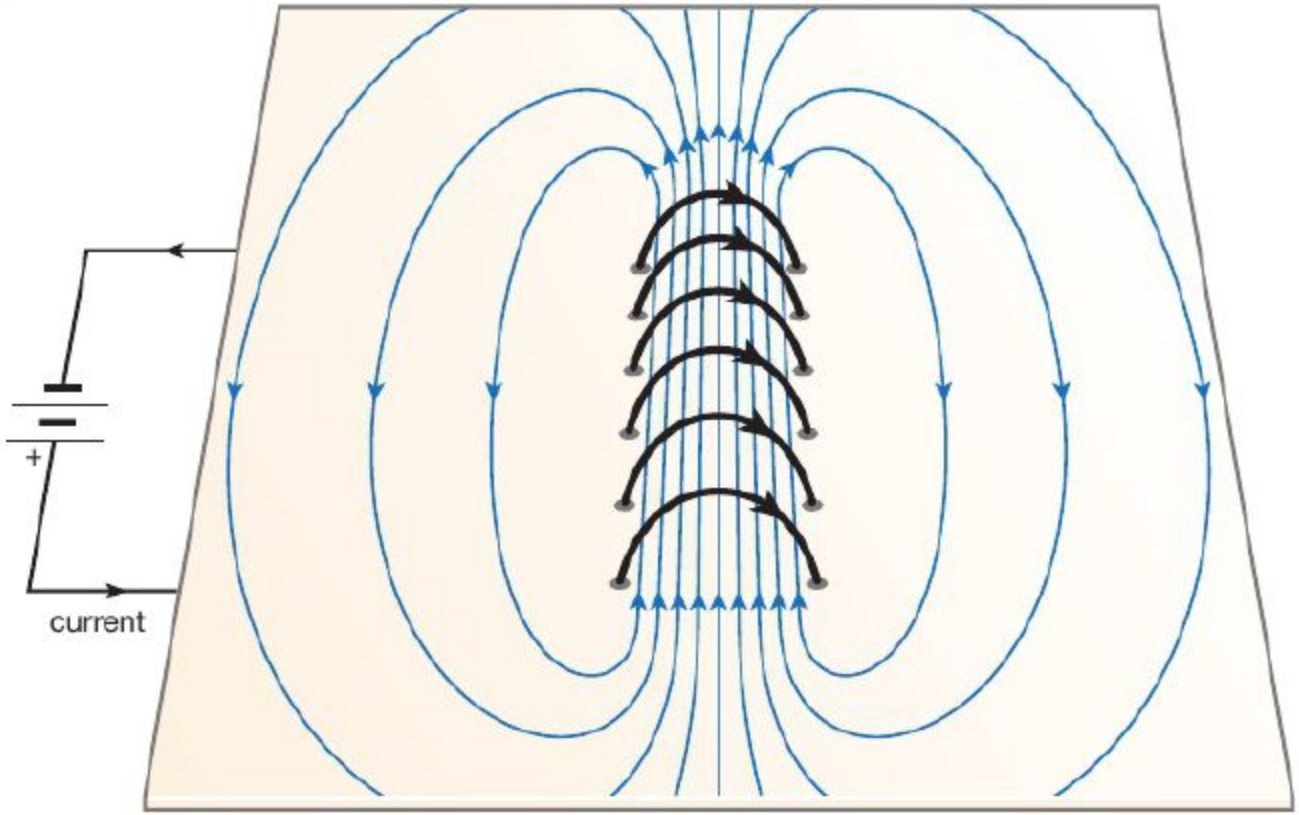
You can work out the direction of the field using the right-hand grip rule (for fields).

If the wire is made into a flat, single-turn coil (circular wire), the magnetic field around the wire changes shape and is as shown in Figure 20.11.

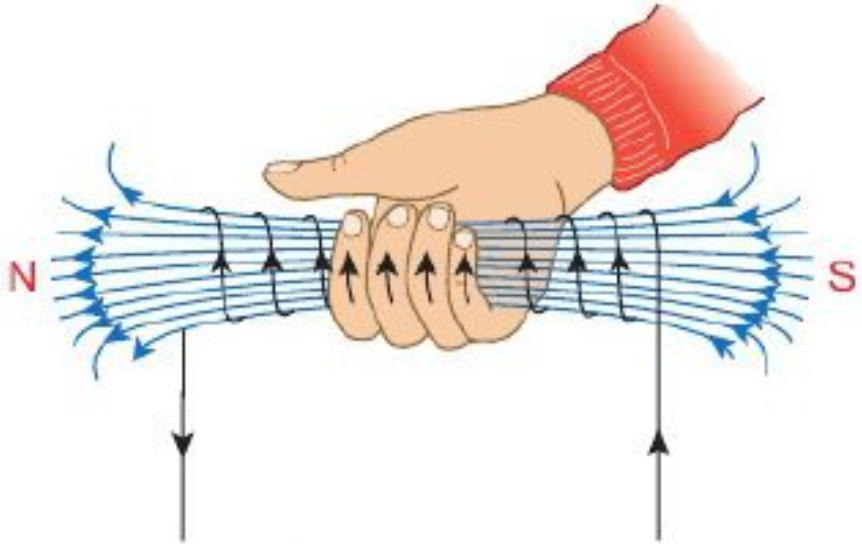


The strength of the magnetic field around a current-carrying wire can be increased by:

1. increasing the current in the wire
2. wrapping the wire into a coil or **solenoid** (a solenoid is a long coil).



The shape of the magnetic field around a solenoid is the same as that around a bar magnet.

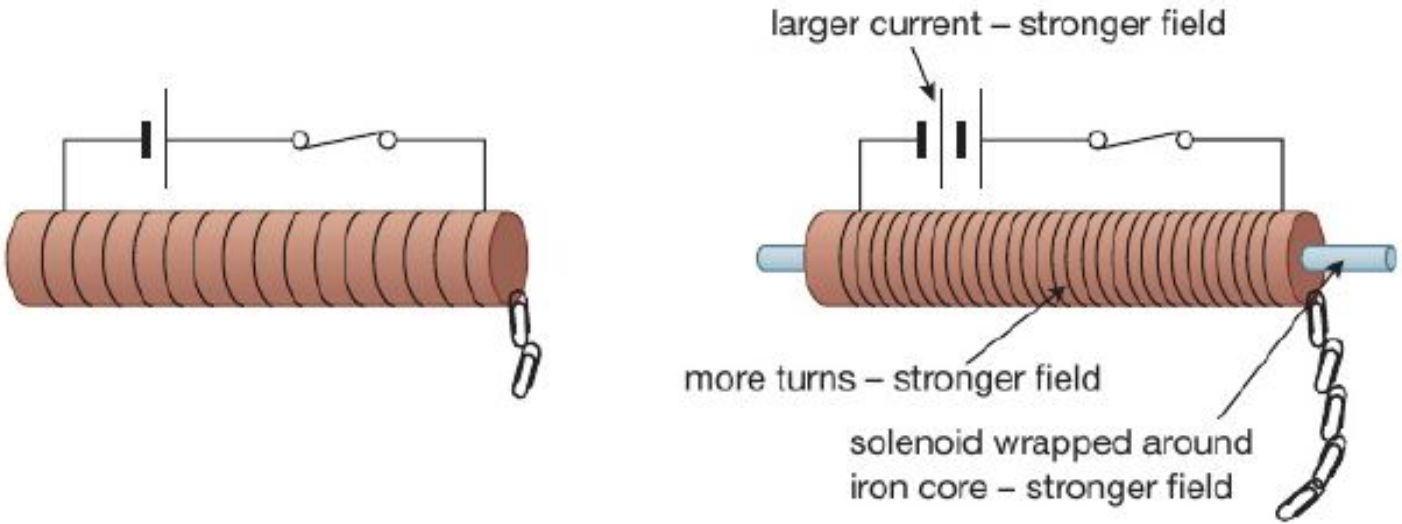


▲ Figure 20.13 You can work out the **polarity** of the solenoid by imagining that your right hand is wrapped around it. Your fingers point in the direction of the current and your thumb points to the north pole of the solenoid. This is the right-hand grip rule (for poles).

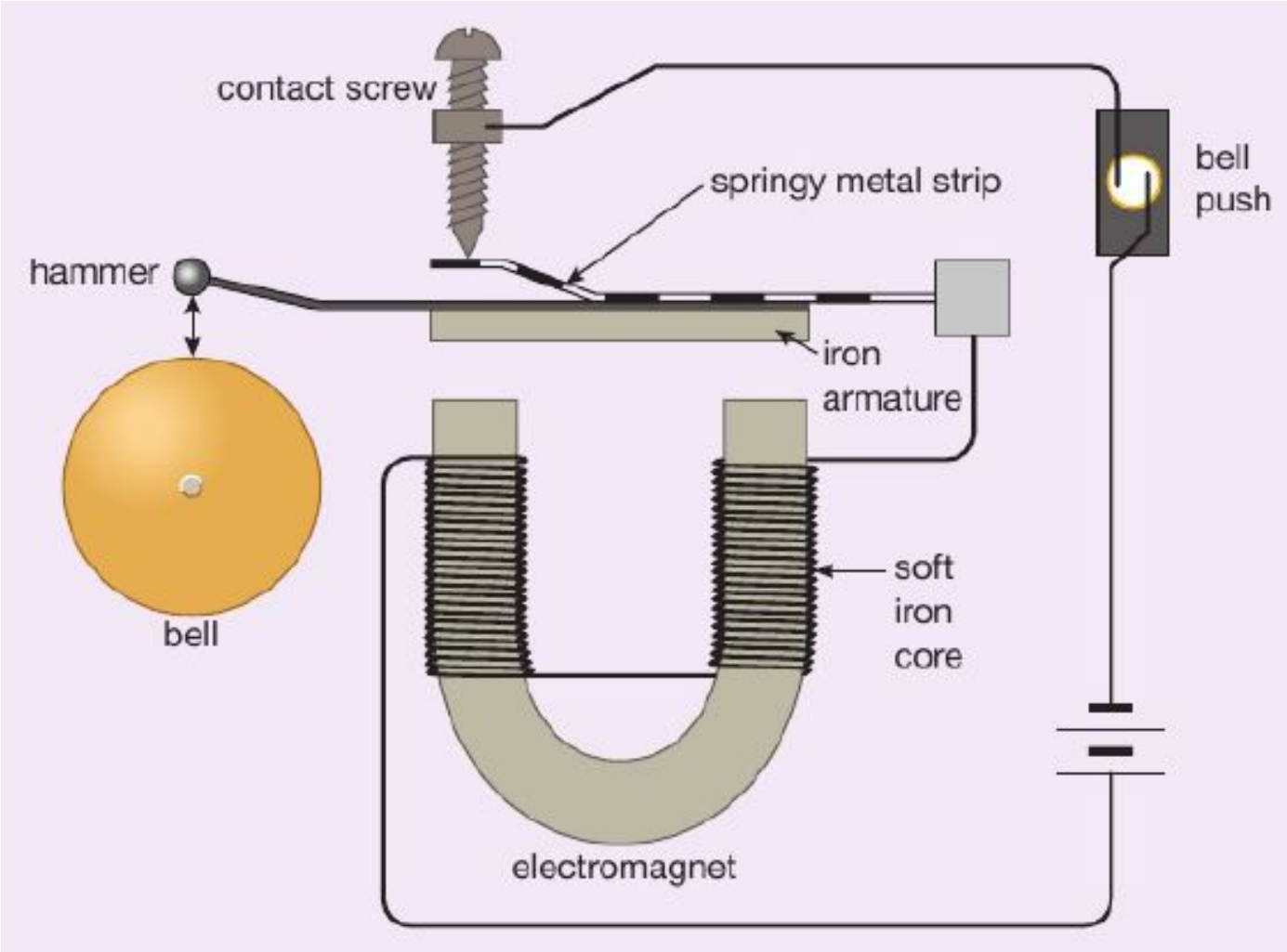
The strength of the field around a solenoid can be increased by:

1. increasing the current in the solenoid
2. increasing the number of turns on the solenoid
3. wrapping the solenoid around a magnetically soft core such as iron.

This combination of soft iron core and solenoid is often referred to as an electromagnet.

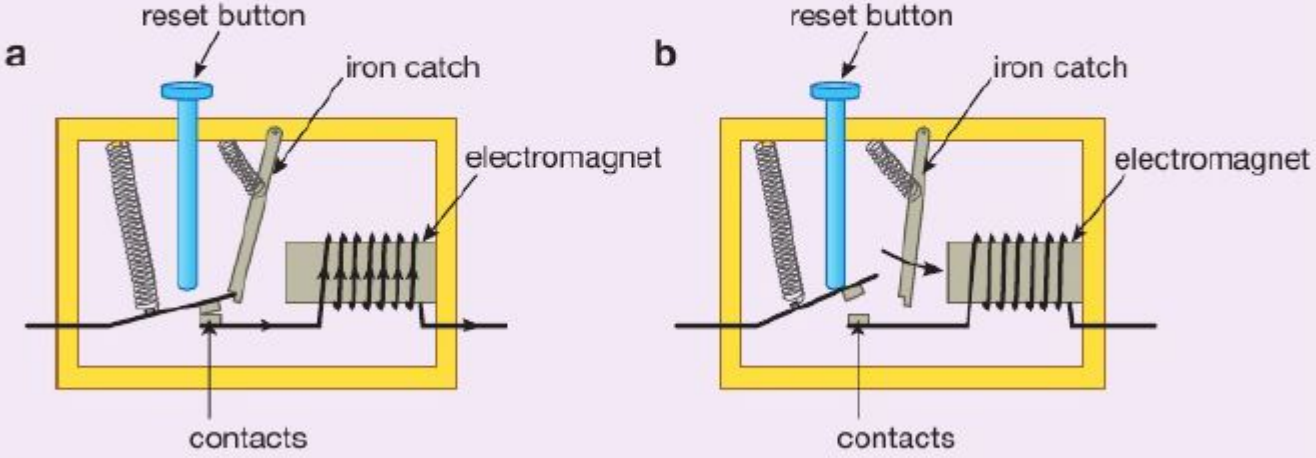


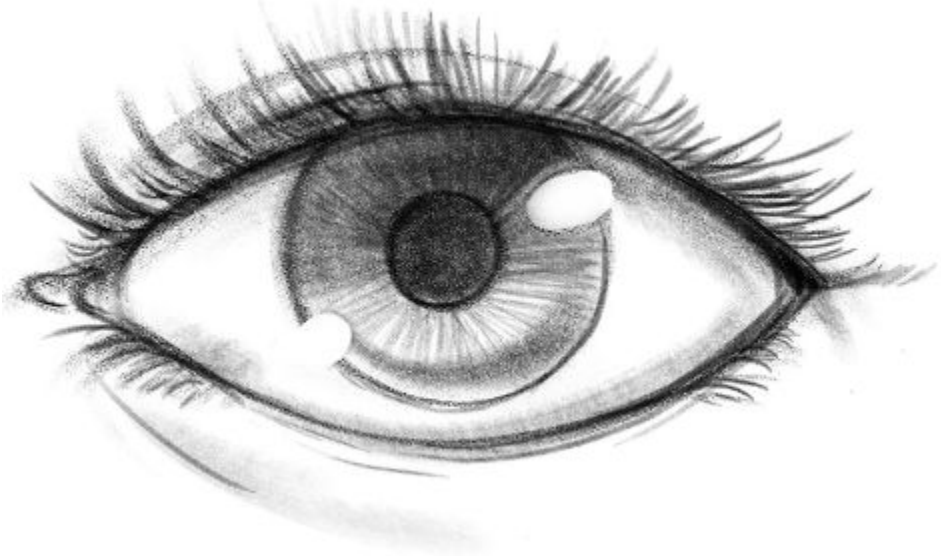
▲ Figure 20.14 Altering the strength of the field around an electromagnet


contact screw
springy metal strip
bell
push
hammer
iron
armature
bell
soft
iron
core
electromagnet

Circuit breaker

Circuit breakers also use electromagnets. The circuit breaker in Figure 20.16 uses electromagnetism to cut off the current if it becomes larger than a certain value. If the current is too high the electromagnet becomes strong enough to pull the iron catch out of position so that the contacts open and the circuit breaks. Once the problem in the circuit has been corrected the catch is repositioned by pressing the reset button.



▲ Figure 20.16 **a** The circuit breaker allows a normal sized current. **b** When there is too large a current the circuit becomes incomplete.

Good Luck & Happy Reading